FOREWORD TO THE 4TH EDITION

Nuclear chemistry is one of the oldest nuclear sciences. It may be defined as the use of nuclear properties to study chemical phenomena, and the use of chemistry to study nuclear properties. However, the topics covered in this book are considerably wider.

The chemical behaviour of the heavy radioactive elements has lead to Nobel prices and international fame for many great scientists since the late 19th century. From the beginning the use of radiation was used for the good of humanity as treatment for different diseases, real or imagined. However, as will all new discoveries, many odd products such as e.g. radium soap hit the market and today we know that such use of radium is not healthy. However, the new knowledge also opened the possibility to use radioactive elements for the treatment of cancerous micro metastases by grafting them on antibodies seeking out the cancer. As the knowledge of nuclear chemistry advanced more elements were discovered. Today the periodic table is still expanding with new element as a result of the nuclear research. The powers of the nuclear forces have been used for both destruction and for building up peaceful power production for the good of society. In the future we expect that nuclear power will be made considerably more sustainable by a significantly better utilization of the uranium resource by the recycling of used nuclear fuel.

Seen in this context nuclear chemistry has still a significant part to play in the future development of a peaceful, sustainable society. Thus also the teaching on different levels will need to learn, not only the basic concepts of the atom, nuclear forces, detection and other basic facts but also look at which applications has been and will be used in the nuclear area.

The history of the present book dates back to 1964 when prof. Jan Rydberg first published a book in Swedish about nuclear chemistry to be used for teaching purposes. Later, this Swedish book was extended and translated to English together with prof. Gregory Choppin and the first edition of the current series was made. The present book is the 4th edition of the original Choppin–Rydberg book revised by Liljenzin and Ekberg. However, read and approved by Rydberg and Choppin. Essentially not much has been changed in the more basic chapters since few new facts have been discovered. However, the logic of the chapters have been made more closely to the first edition. The more applied chapters have been extensively edited and updated in order to mirror the latest discoveries in the still evolving areas such as production of new elements, new separation schemes for used nuclear fuel, and the discovery of new elementary particles.

This detailed update would not have been possible without consultation with experts in the different fields covered by the book. Especially we would like to express our

gratitude to prof. Gunnar Skarnemark, prof. Henrik Ramebäck, prof. Henrik Nylén, prof. Janne Wallenius, and to all those who have helped us to improve the mauscript and to check the proofs.

We hope that his new edition will be an aid in the teaching of nuclear chemistry and also be a good companion after the education whenever some facts or equations have to be refreshed.

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